

DSA 4-Day Intensive Study Plan (Java)

Day 1: Arrays, Strings, Hashing, and Complexity

- Morning (3–4 hours): Learn Big O notation ($O(1)$, $O(\log n)$, $O(n)$, $O(n^2)$). Arrays: traversal, min/max, sum, second largest, reverse array. Practice: largest element, second largest element, move zeros, reverse array.
- Afternoon (3–4 hours): Strings: reverse string, palindrome, count vowels, character frequency. HashMap: frequency counting, lookup optimization. Practice: first non-repeating character, valid anagram, Two Sum.
- Evening (2 hours): Solve 5–8 easy problems without looking at solutions.

Day 2: Recursion, Sorting, Binary Search

- Morning: Recursion fundamentals, factorial, Fibonacci, sum of N numbers, print 1 to N.
- Afternoon: Bubble Sort, Selection Sort, Insertion Sort. Understand time complexity and implement each from scratch.
- Evening: Binary Search, first/last occurrence, square root, search insert position.

Day 3: Linked List, Stack, Queue

- Morning: Singly Linked List, insert, delete, traverse. Practice insertions and deletions.
- Afternoon: Stack operations (push, pop, peek), balanced parentheses, reverse string using stack. Queue operations and implementation.
- Evening: Solve 5–6 mixed problems.

Day 4: Trees, Heap, Graphs

- Morning: Binary Tree traversals (DFS, BFS, preorder, inorder, postorder).
- Afternoon: Heap basics, PriorityQueue in Java, K largest elements. Graph basics: DFS and BFS.
- Evening: Full revision of all major topics.

Daily Problem Solving Routine

- Read problem (2 min)
- Solve manually on paper (5 min)
- Write pseudocode (5 min)
- Code in Java (10–15 min)
- Analyze complexity (2 min)
- Optimize if possible (5 min)

Must-Solve Problems

- Largest Element, Second Largest Element, Reverse Array, Move Zeroes, Two Sum, Valid Anagram, Palindrome String, Binary Search, Factorial, Fibonacci, Reverse Linked List, Valid Parentheses, Tree Traversals, BFS, DFS.